



Report No. 617

**Gravimetric
Proficiency Testing Program**

Round One

July 2009

ACKNOWLEDGMENTS

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1. **FOREWORD**

This report summarises the results of the interlaboratory comparison program Gravimetric Round One.

The exercise was conducted during the period October 2008 to June 2009 by Proficiency Testing Australia. The Program Coordinator was Dr M Li. The main aim of the program was to assess laboratories' ability to competently determine the weight of two aluminium weights (nominally 5 mg and 20 mg respectively).

2. **STATISTICAL DESIGN OF THE PROGRAM**

Five sets of weights (A, B, C, D and E) were distributed amongst the 21 participating laboratories. Each set consisted of two aluminium weights, one nominally 5 mg, and the other nominally 20 mg. The samples were originally labelled A05, A20, B05, B20, C05, C20, D05, D20, E05 and E20, however, each specimen was re-labelled with a unique sample number between participants.

The program was based on a measurement comparison design, where the artefacts (two aluminium weights) were sequentially distributed around the 21 participants. Five sets of weights (A, B, C, D and E) were used, to reduce the time required to cover all 21 laboratories. The Standards Laboratory of Measurement & Technical Services, NSW Office of Fair Trading acted as the reference laboratory, deriving the reference values at the beginning and the end of the program.

A summary of results returned by the participating laboratories, compared to the reference values, appears in Appendix A1 to A5. Measurement performance is judged on the basis of an E_n number for each measurement. The E_n number is an internationally accepted method for determining the agreement of individual results with the reference values in relation to the uncertainties of measurement of each. That is, the E_n number indicates whether laboratories are within their particular uncertainty of measurement of the reference value. The E_n ratio is defined as:

$$E_n = \frac{\text{Lab Result} - \text{Ref Value}}{\sqrt{(U_{95} \text{ Lab})^2 + (U_{95} \text{ Ref})^2}}$$

where U_{95} is the reported uncertainty of measurement at a 95% confidence level. For the results to be acceptable, values of $|E_n|$ less than one are required.

3. **FEATURES OF THE PROGRAM**

- (a) A total of 21 laboratories received samples. Set A was distributed to participants with laboratory codes 2, 7, 12 and 17, Set B was distributed to participants with laboratory codes 3 and 8, Set C was distributed to participants with laboratory codes 4, 9, 14 and 19, Set D was distributed to participants with laboratory codes 5, 10, 15, 18 and 20, while Set E was distributed to participants with laboratory codes 1, 6, 11, 13, 16 and 21. A 5 mg weight from Set C was lost during distribution.

- (b) Participants were supplied two aluminium weights, one nominally 5 mg, the other nominally 20 mg.
- (c) Prior to sample distribution, and again at the conclusion of the round, all weights (except the 5 mg weight in Set C) were calibrated by the Standards Laboratory of Measurement & Technical Services, NSW Office of Fair Trading. The 5 mg weight in Set C was only calibrated prior to sample distribution since it was lost.
- (d) The results for each sample set as reported by participants are presented in Appendix A, together with calculated E_n numbers (A1- A5) and graphical presentations of the data (A7). A table of participant's reported uncertainties of measurement is also presented in Appendix A6.
- (e) Participating laboratories were requested to perform the tests according to the "Instructions to Participants", and to record their results on the accompanying "Results Sheet", all of which were distributed to laboratories with the weights. Copies of the "Instructions to Participants" and "Results Sheet" are given in Appendix B of this report.
- (f) Each laboratory was randomly allocated a unique code number for the program to ensure confidentiality of results. Reference to each laboratory in this report is by its code number. Please note that some laboratories reported more than one set of results, therefore, one code number (with letter) could appear several times in the same date set.

4. **FORMAT OF APPENDICES**

Appendix A

For each sample set (A to E), the following information is provided for each sample.

- (i) The weight determination (in mg) as reported by participating laboratories, together with their reported uncertainties of measurement at the 95% confidence level (U_{95}).
- (ii) The participating laboratories' results and the reference value in mg.
- (iii) The calculated E_n number for each participant's weight determinations.
- (iv) A graph for each sample, displaying each participant's value, represented by a black diamond, together with their reported uncertainty of measurement, represented by bars extending above and below its value (see Appendix A7).

Additional information provided in Appendix A is:

- (i) A table containing the participating laboratories' reported uncertainties of measurement, together with the median, minimum and maximum reported uncertainty.

- (ii) A table (Appendix A8) containing additional information provided by participants in relation to the microbalances used.

Appendix B

- (i) Instructions to Participants
- (ii) Results Sheet

5. **OUTLIER RESULTS**

In order to achieve the program's aim of assessing laboratories' testing performance, E_n numbers have been calculated for each participant's weight determination. The E_n number indicates whether laboratories are within their particular uncertainty of measurement of the reference value. For the result to be considered acceptable, the E_n number must lie between -1 and $+1$ (i.e. $|E_n| < 1$). For laboratory codes 10a and 10b, the reported uncertainties were not able to be used to calculate their E_n numbers.

6. **PTA AND TECHNICAL ADVISER'S COMMENTS**

Background Information

This interlaboratory comparison program was designed to assist laboratories that perform the determination of inhalable/respirable dust, welding fumes and gases according to the following Australian Standards: AS2985-2004^[2], AS3640-2004^[3] and AS3853.1-2006^[4].

It is acknowledged that humidity and electrostatic charges are a major source of uncertainty in these analyses, and considerable thought was put into trying to include these factors in the scheme. These factors, however, could not be included in such a program, and can only be addressed by appropriate analytical methodology and correct laboratory practice. This can only be addressed during laboratory assessments.

Informal round robins conducted in Australia and UK some years ago indicated that some laboratories using calibrated weights and calibrated balances obtained grossly incorrect answers. Additionally, laboratory technique can provide further random and/or systematic errors. The present program is the formal attempt to use a simple test for the above errors, and then to assist outlier laboratories overcome their problems.

For such a simple program, the overall findings were remarkable, with deficiencies identified despite the basic nature of the program.

Uncertainties of Measurement (U_{95})

The uncertainties of measurement reported by some laboratories appear too small. Conversely, the U_{95} values reported by some laboratories appear too large. These laboratories should review their uncertainty estimates.

For laboratory codes 16a and 16b, the reported uncertainties were not able to be used to calculate their E_n numbers. A number of laboratories did not report the range of microbalance as requested by PTA. The microbalance range should be provided to allow a comprehensive review to occur.

It should be noted that some laboratories who reported large uncertainties have been classed as satisfactory in terms of their E_n number, when in fact they may have been identified as an outlier had they reported a more appropriate uncertainty. Alternatively, participants reporting small uncertainties may have falsely excluded themselves from being classed as satisfactory. This should be taken into account when assessing participants performance in this program.

Possible Sources of Error

As mentioned previously, the use of five place balances is a major source of error in this program, due to the problem of conforming with the current Australian standard balance requirements. Weight calibration certificates may also be another source of error, as some are issued with very large and inappropriate uncertainties that should not be applied to microbalances.

Some laboratories submitting results for more than one analyst were identified with outlying results for one analyst and not the others. This could be due to inadequate training or erratic balances. Furthermore, some analysts submitted acceptable results for the 20 mg weight, but not for the 5 mg weight (and vice versa), which may indicate balance or weight calibration problems.

The stainless steel reference weights are suggested to be used in the future program.

7. REFERENCES

- [1] Guide to Proficiency Testing Australia (2008). (This document is located on the PTA website at www.pta.asn.au under Programs / Documents).
- [2] AS2985-2004: *'Workplace atmospheres - Method for sampling and gravimetric determination of respirable dust'*.
- [3] AS3640-2004: *'Workplace atmospheres – Method for sampling and gravimetric determination of inhalable dust'*.
- [4] AS3853.1-2006: *'Health and safety in welding and allied processes - Sampling of airborne particles and gases in the operator's breathing zone - Sampling of airborne particles'*.

APPENDIX A

Summary of Results

Results for Set A	A1.1
Results for Set B	A2.1
Results for Set C	A3.1
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Results for Set E	A5.1
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Graphical Presentations	A7.1
Additional Information Provided	A8.1

A1.1

Appendix A1

SAMPLES A05 and A20

Lab Code	Nominated 5/6 place	Nominal 5 mg (A05)					Nominal 20 mg (A20)				
		Sample No.	Weight (mg)	U ₉₅ (mg)	LAB - REF (mg)	E _n	Sample No.	Weight (mg)	U ₉₅ (mg)	LAB - REF (mg)	E _n
2	5	1a	5.17	0.3	-0.019	-0.06	2a	19.75	0.3	-0.008	-0.02
7	5	3a	5.19	0.06	0.001	0.02	4a	19.74	0.06	-0.018	-0.29
12a	6	5a	5.18	0.02	-0.009	-0.40	6a	19.75	0.02	-0.008	-0.32
12b	6	5a	5.18	0.02	-0.009	-0.40	6a	19.75	0.02	-0.008	-0.32
17	6	7a	5.189	0.005	0.000	0.00	8a	19.761	0.005	0.003	0.26

Reference Lab Initial: 5.189 0.007

19.757 0.009

Reference Lab Final: 5.189 0.007

19.758 0.009

Average: **5.189** **0.010** (Expanded Uncertainty)**19.758** **0.013** (Expanded Uncertainty)

NOTES:

1. The current Australian Standards now accept 5 and 6 place balances which give very different uncertainties. These different uncertainties must be taken into account in terms of the significance of results generated by each class of balance.
2. E_n refers to Error Normalised. An E_n number between -1 and +1 is considered acceptable (i.e. |E_n| < 1).
3. U₉₅ refers to uncertainty of measurement at the 95% confidence level.
4. Weight and U₉₅ in the above table are shown 'as reported' by participants.
5. LAB - REF refers to Participating Laboratory's result minus the Reference Laboratory's result.
6. Samples A05 (nominal 5 mg) and A20 (nominal 20 mg) were relabelled between participants (unique sample numbering shown above).
7. Where a laboratory submitted results for more than one analyst, an 'a', 'b' etc has been added to the numerical code to denote the different analyst.
8. The expanded uncertainty of measurement for the Reference Laboratory takes into account the variation between the initial and final calibration, and represents the true uncertainty of measurement at the 95% confidence level.

A2.1

Appendix A2

SAMPLES B05 and B20

Lab Code	Nominated 5/6 place	Nominal 5 mg (B05)					Nominal 20 mg (B20)				
		Sample No.	Weight (mg)	U ₉₅ (mg)	LAB - REF (mg)	E _n	Sample No.	Weight (mg)	U ₉₅ (mg)	LAB - REF (mg)	E _n
3a	5	9b	5.246	0.020	0.002	0.09	10b	19.872	0.020	0.003	0.13
3b	5	9b	5.26	0.02	0.016	0.76	10b	19.91	0.02	0.041	1.73
3c	5	9b	5.251	0.020	0.007	0.33	10b	19.871	0.020	0.002	0.08
8	5	11b	5.24	0.0257	-0.004	-0.15	12b	19.86	0.0257	-0.009	-0.31

Reference Lab Initial: 5.244 0.007

19.869 0.009

Reference Lab Final:

19.869 0.009

Average: **5.244** **0.007** (Expanded Uncertainty)**19.869** **0.013** (Expanded Uncertainty)

NOTES:

1. The current Australian Standards now accept 5 and 6 place balances which give very different uncertainties. These different uncertainties must be taken into account in terms of the significance of results generated by each class of balance.
2. E_n refers to Error Normalised. An E_n number between -1 and +1 is considered acceptable (i.e |E_n| < 1).
3. U₉₅ refers to uncertainty of measurement at the 95% confidence level.
4. Weight and U₉₅ in the above table are shown 'as reported' by participants.
5. LAB - REF refers to Participating Laboratory's result minus the Reference Laboratory's result.
6. Samples B05 (nominal 5 mg) and B20 (nominal 20 mg) were relabelled between participants (unique sample numbering shown above).
7. Where a laboratory submitted results for more than one analyst, an 'a', 'b' etc has been added to the numerical code to denote the different analyst.
8. The expanded uncertainty of measurement for the Reference Laboratory takes into account the variation between the initial and final calibration, and represents the true uncertainty of measurement at the 95% confidence level.

A3.1

Appendix A3

SAMPLES C05 and C20

Lab Code	Nominated 5/6 place	Nominal 5 mg (C05)					Nominal 20 mg (C20)				
		Sample No.	Weight (mg)	U ₉₅ (mg)	LAB - REF (mg)	E _n	Sample No.	Weight (mg)	U ₉₅ (mg)	LAB - REF (mg)	E _n
4	5	17c	5.25	0.05	0.008	0.15	18c	19.53	0.08	0.000	-0.01
9	5	19c	0.00525	0.7	-5.237	-7.48	20c	0.01951	0.2	-19.511	-97.36
14	6	21c	5.245	0.009	0.003	0.19	22c	19.525	0.009	-0.006	-0.35
19	6	23c	5.248	0.002	0.006	0.54	24c	19.526	0.005	-0.005	-0.33

Reference Lab Initial: 5.243 0.007

19.527 0.009

Reference Lab Final: 5.242 0.007

19.534 0.009

Average: **5.243** **0.010** (Expanded Uncertainty)**19.531** **0.013** (Expanded Uncertainty)

NOTES:

1. The current Australian Standards now accept 5 and 6 place balances which give very different uncertainties. These different uncertainties must be taken into account in terms of the significance of results generated by each class of balance.
2. E_n refers to Error Normalised. An E_n number between -1 and +1 is considered acceptable (i.e |E_n| < 1).
3. U₉₅ refers to uncertainty of measurement at the 95% confidence level.
4. Weight and U₉₅ in the above table are shown 'as reported' by participants.
5. LAB - REF refers to Participating Laboratory's result minus the Reference Laboratory's result.
6. Samples C05 (nominal 5 mg) and C20 (nominal 20 mg) were relabelled between participants (unique sample numbering shown above).
7. The expanded uncertainty of measurement for the Reference Laboratory takes into account the variation between the initial and final calibration, and represents the true uncertainty of measurement at the 95% confidence level.

A4.1

Appendix A4

SAMPLES D05 and D20

Lab Code	Nominated 5/6 place	Nominal 5 mg (D05)					Nominal 20 mg (D20)				
		Sample No.	Weight (mg)	U ₉₅ (mg)	LAB - REF (mg)	E _n	Sample No.	Weight (mg)	U ₉₅ (mg)	LAB - REF (mg)	E _n
5a	5	25d	5.24	0.05	0.003	0.05	26d	19.55	0.08	0.011	0.14
5b	5	25d	5.24	0.05	0.003	0.05	26d	19.53	0.08	-0.009	-0.11
5c	5	25d	5.22	0.05	-0.018	-0.34	26d	19.49	0.08	-0.049	-0.60
10	5	27d	5.24	0.016	0.003	0.13	28d	19.54	0.016	0.001	0.05
15	6	29d	5.2474	0.0045	0.010	0.86	30d	19.5513	0.0045	0.012	0.91
18	6	15d	5.245	0.002	0.008	0.69	16d	19.538	0.005	-0.001	-0.07
20	6	31d	5.252	0.001	0.015	1.36	32d	19.543	0.001	0.004	0.31

Reference Lab Initial: 5.234 0.008

19.540 0.009

Reference Lab Final: 5.241 0.007

19.538 0.009

Average: **5.238 0.011 (Expanded Uncertainty)****19.539 0.013 (Expanded Uncertainty)**

NOTES:

1. The current Australian Standards now accept 5 and 6 place balances which give very different uncertainties. These different uncertainties must be taken into account in terms of the significance of results generated by each class of balance.
2. E_n refers to Error Normalised. An E_n number between -1 and +1 is considered acceptable (i.e. |E_n| < 1).
3. U₉₅ refers to uncertainty of measurement at the 95% confidence level.
4. Weight and U₉₅ in the above table are shown 'as reported' by participants.
5. LAB - REF refers to Participating Laboratory's result minus the Reference Laboratory's result.
6. Samples D05 (nominal 5 mg) and D20 (nominal 20 mg) were relabelled between participants (unique sample numbering shown above).
7. Where a laboratory submitted results for more than one analyst, an 'a', 'b' etc has been added to the numerical code to denote the different analyst.
8. The expanded uncertainty of measurement for the Reference Laboratory takes into account the variation between the initial and final calibration, and represents the true uncertainty of measurement at the 95% confidence level.

A5.1

Appendix A5

SAMPLES E05 and E20

Lab Code	Nominated 5/6 place	Nominal 5 mg (E05)					Nominal 20 mg (E20)				
		Sample No.	Weight (mg)	U ₉₅ (mg)	LAB - REF (mg)	E _n	Sample No.	Weight (mg)	U ₉₅ (mg)	LAB - REF (mg)	E _n
1	5	41e	5.20	0.01	-0.005	-0.32	42e	19.90	0.01	0.016	0.97
6	5	33e	5.20	0.04	-0.005	-0.11	34e	19.88	0.04	-0.004	-0.10
11a	6	35e	5.207	0.015	0.002	0.13	36e	19.882	0.015	-0.002	-0.12
11b	6	35e	5.208	0.015	0.003	0.18	36e	19.881	0.015	-0.003	-0.17
11c	6	35e	5.210	0.015	0.005	0.29	36e	19.883	0.015	-0.001	-0.07
11d	6	35e	5.208	0.015	0.003	0.18	36e	19.885	0.015	0.001	0.03
11e	6	35e	5.205	0.015	0.000	0.02	36e	19.883	0.015	-0.001	-0.07
13	6	13e	5.204	0.007	-0.001	-0.05	14e	19.877	0.009	-0.007	-0.47
16a	6	37e	5.204	0.05xC*	-0.001	N/A	38d	19.883	0.05xC*	0.003	N/A
16b	6	37e	5.204	0.05xC*	-0.001	N/A	38d	19.882	0.05xC*	0.002	N/A
21	6	39e	5.2047	0.015	0.000	-0.02	40e	19.88435	0.015	0.004	0.22

Reference Lab Initial: 5.204 0.008

Reference Lab Final: 5.206 0.007

Average: **5.205** **0.011** (Expanded Uncertainty)

19.880 0.009

19.880 0.009

19.880 **0.013** (Expanded Uncertainty)

NOTES:

1. The current Australian Standards now accept 5 and 6 place balances which give very different uncertainties. These different uncertainties must be taken into account in terms of the significance of results generated by each class of balance.
2. E_n refers to Error Normalised. An E_n number between -1 and +1 is considered acceptable (i.e. |E_n| < 1).
3. U₉₅ refers to uncertainty of measurement at the 95% confidence level.
4. Weight and U₉₅ in the above table are shown 'as reported' by participants.
5. LAB - REF refers to Participating Laboratory's result minus the Reference Laboratory's result.
6. Samples E05 (nominal 5 mg) and E20 (nominal 20 mg) were relabelled between participants (unique sample numbering shown above).
7. Where a laboratory submitted results for more than one analyst, an 'a', 'b' etc has been added to the numerical code to denote the different analyst.
8. The expanded uncertainty of measurement for the Reference Laboratory takes into account the variation between the initial and final calibration, and represents the true uncertainty of measurement at the 95% confidence level.
9. For Laboratory codes 16a and 16b, the reported uncertainties were not able to be used to calculate their E_n numbers.

Appendix A6

UNCERTAINTY OF MEASUREMENT (U_{95}) (as reported by participants)

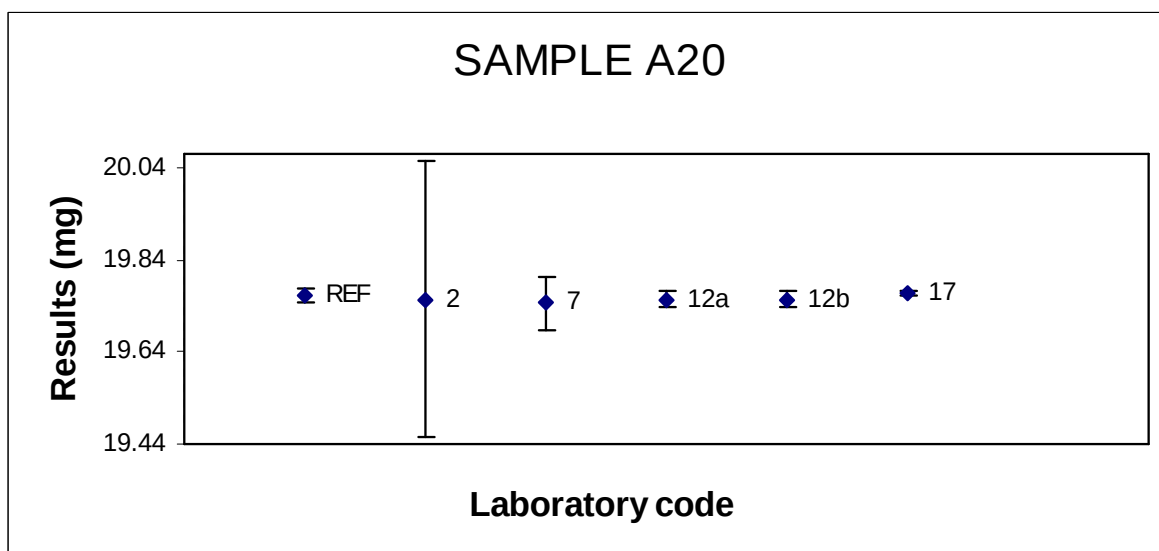
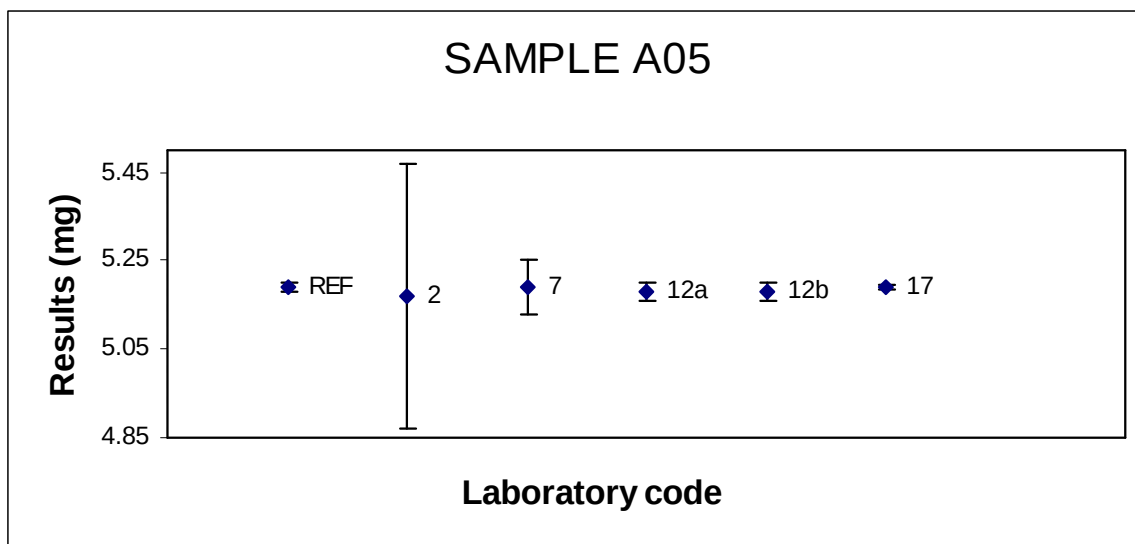
Lab Code	Sample Set	Nominal 5 mg U_{LAB} (mg)	Nominal 20 mg U_{LAB} (mg)
1	E	0.01	0.01
2	A	0.3	0.3
3a	B	0.020	0.020
3b	B	0.02	0.02
3c	B	0.020	0.020
4	C	0.05	0.08
5a	D	0.05	0.08
5b	D	0.05	0.08
5c	D	0.05	0.08
6	E	0.04	0.04
7	A	0.06	0.06
8	B	0.0257	0.0257
9	C	0.7	0.2
10	D	0.016	0.016
11a	E	0.015	0.015
11b	E	0.015	0.015
11c	E	0.015	0.015
11d	E	0.015	0.015
11e	E	0.015	0.015
12a	A	0.02	0.02
12b	A	0.02	0.02
13	E	0.007	0.009
14	C	0.009	0.009
15	D	0.0045	0.0045
16a	E	0.05xC*	0.05xC*
16b	E	0.05xC*	0.05xC*
17	A	0.005	0.005
18	D	0.002	0.005
19	C	0.002	0.005
20	D	0.001	0.001
21	E	0.015	0.015
Median**		0.016	0.016
Minimum		0.001	0.001
Maximum		0.7	0.3

NOTES:

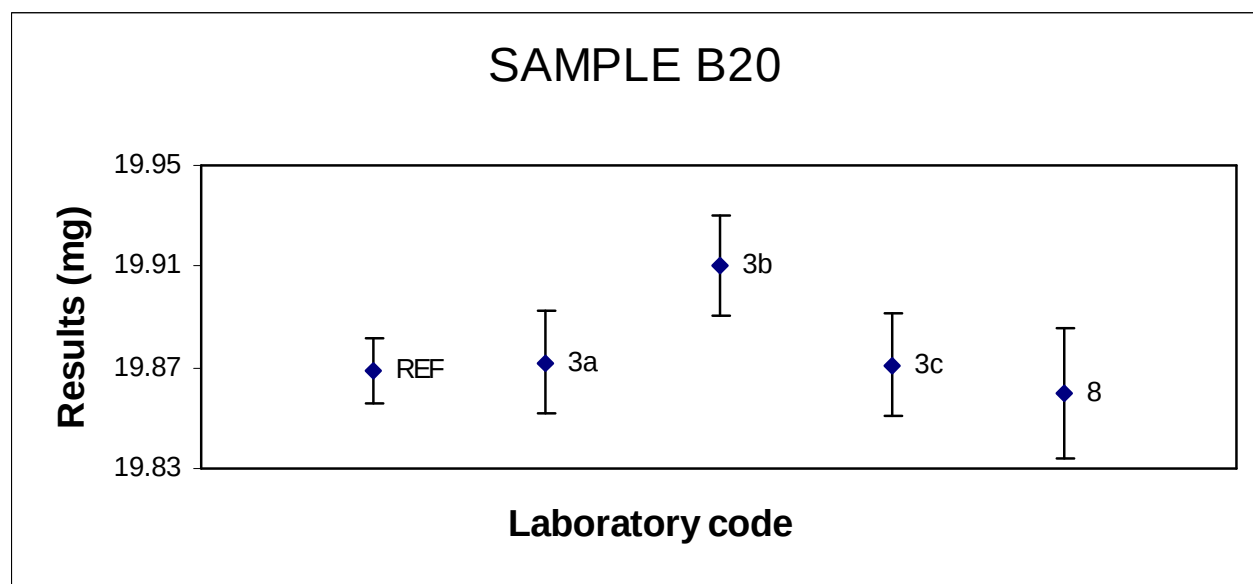
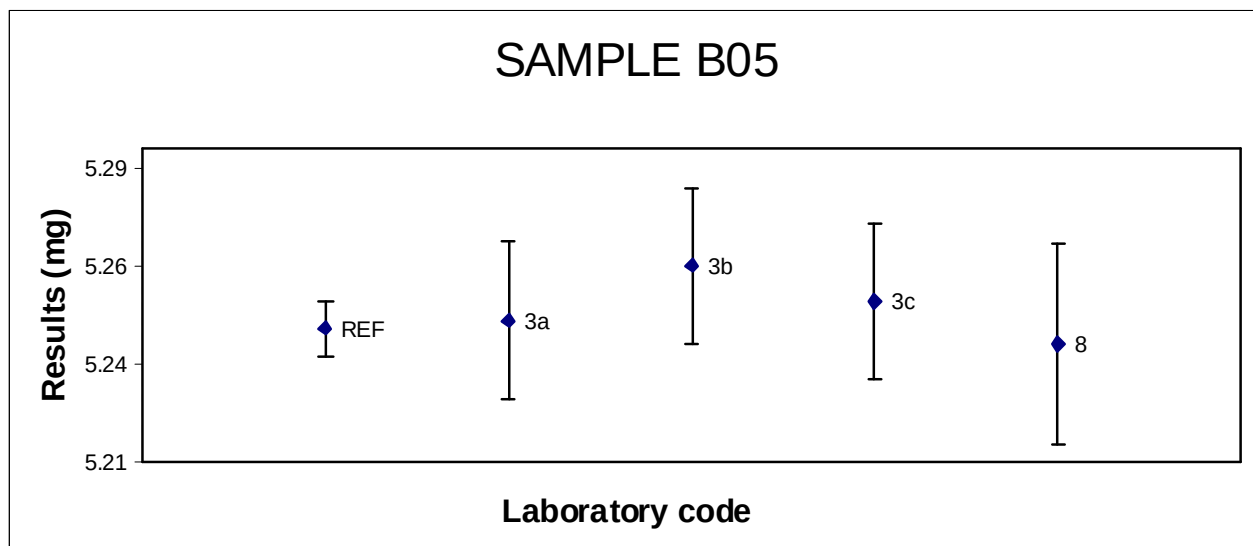
1. Where a laboratory submitted results for more than one analyst, an 'a', 'b' etc has been added to the numerical code to denote the different analyst.
2. U_{95} refers to uncertainty of measurement at the 95% confidence level.
3. The expanded uncertainty of measurement for the Reference Laboratory takes into account the variation between the initial and final calibration.
4. For laboratory codes 10a and 10b, the reported uncertainties were not able to be used to calculate their E_n numbers. C*=dust concentraion (mg/m^3).

Appendix A7

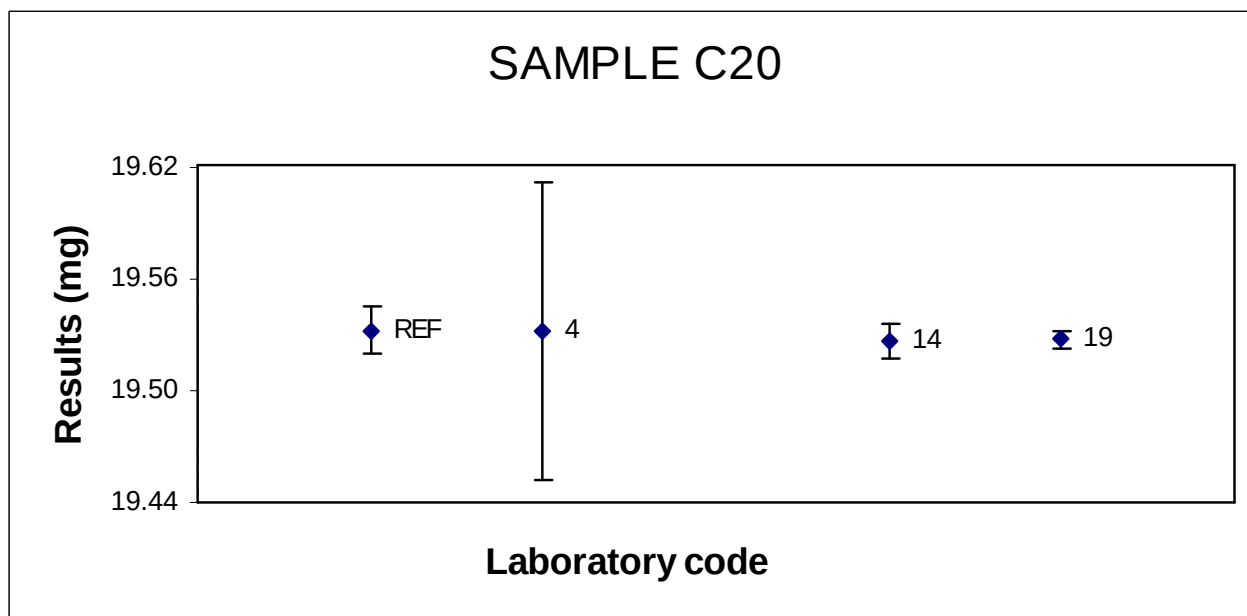
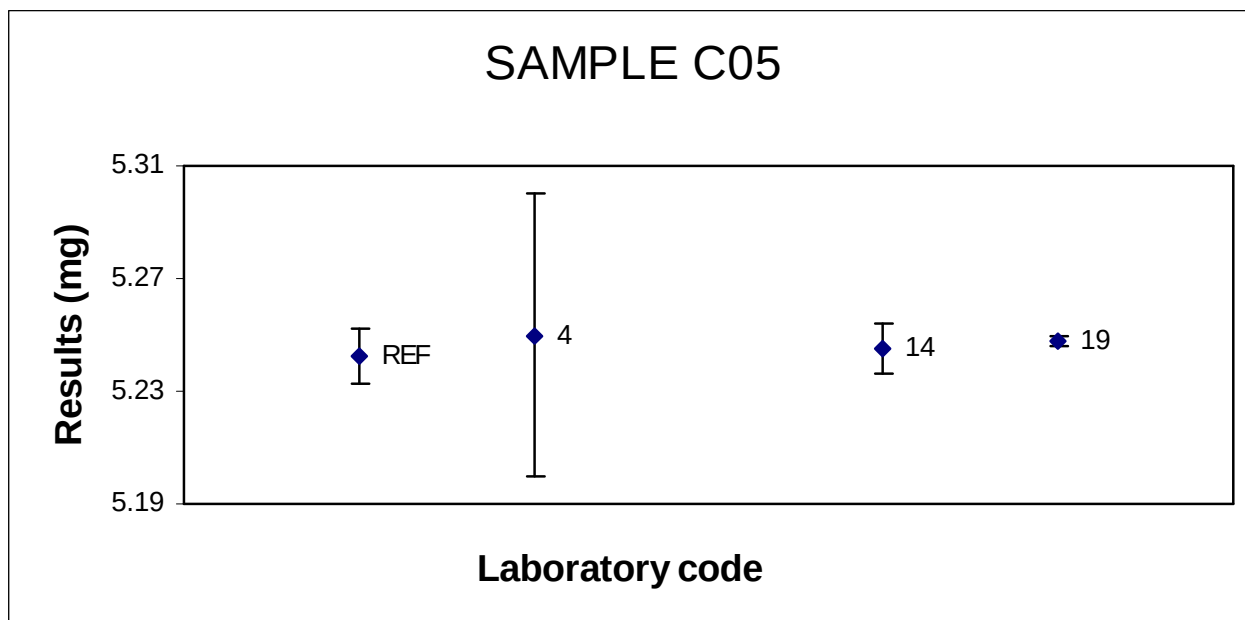
Graphical Presentation of Participant's Results



NOTE: Laboratory codes 2 and 7 nominated 5 place microbalances, and laboratory codes 12a, 12b and 17 nominated 6 place microbalances.

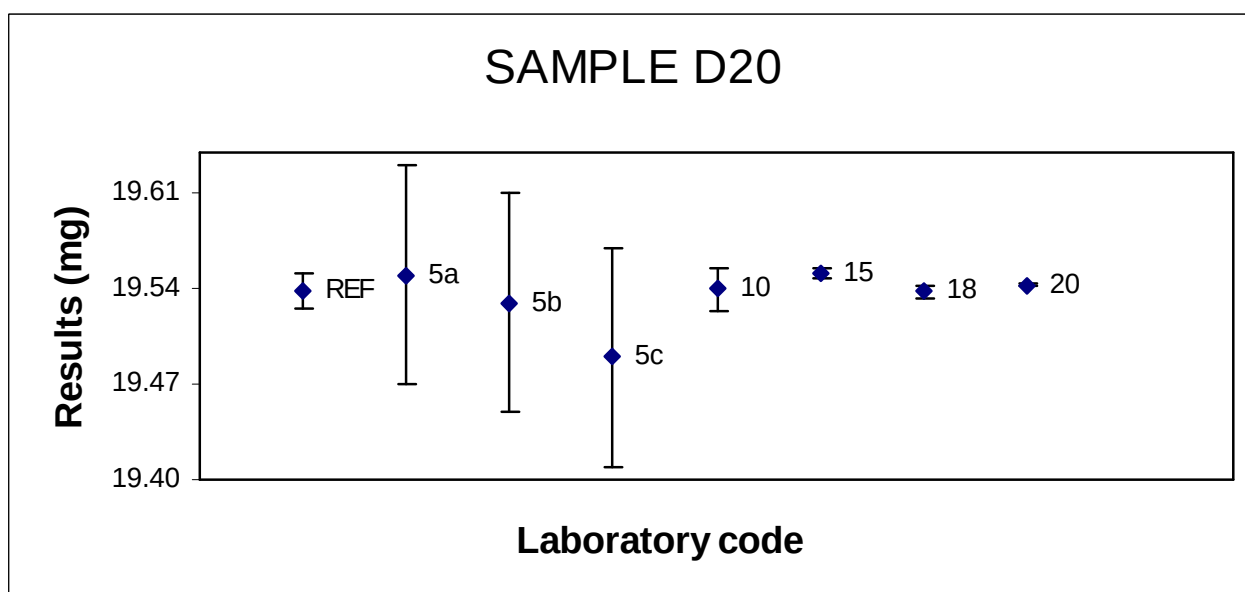
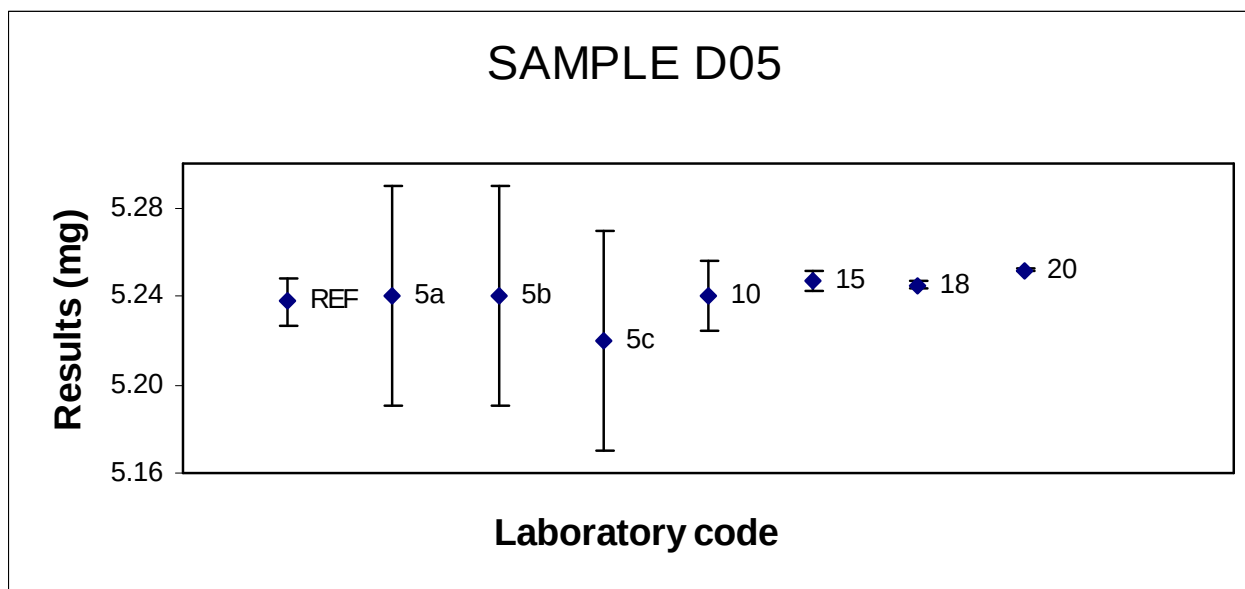
Graphical Presentation of Participant's Results

NOTE: Laboratory codes 3a, 3b, 3c and 8 nominated 5 place microbalances.

Graphical Presentation of Participant's Results

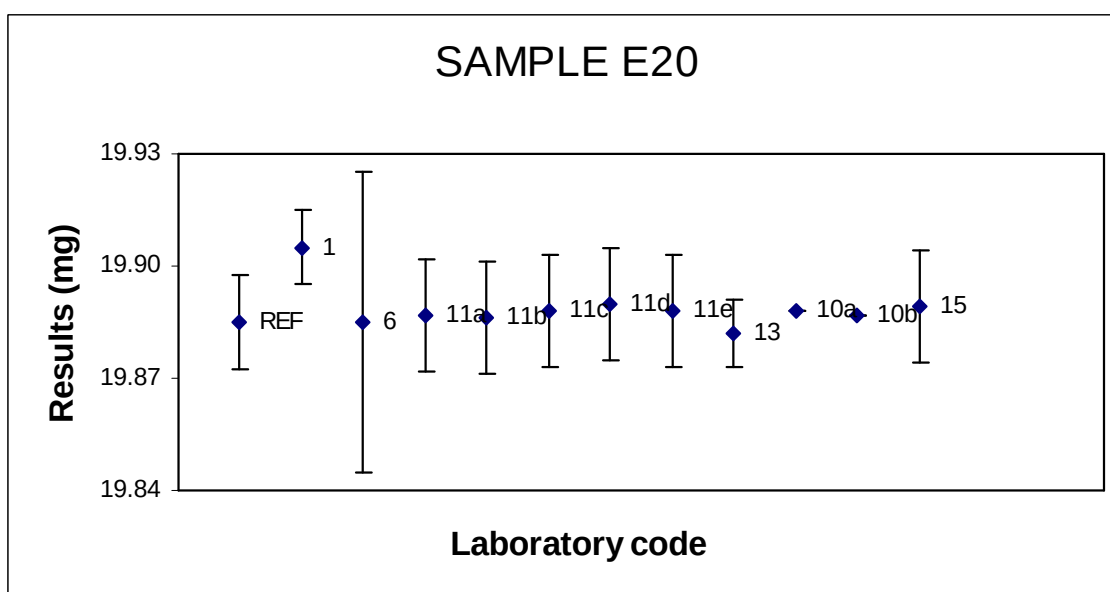
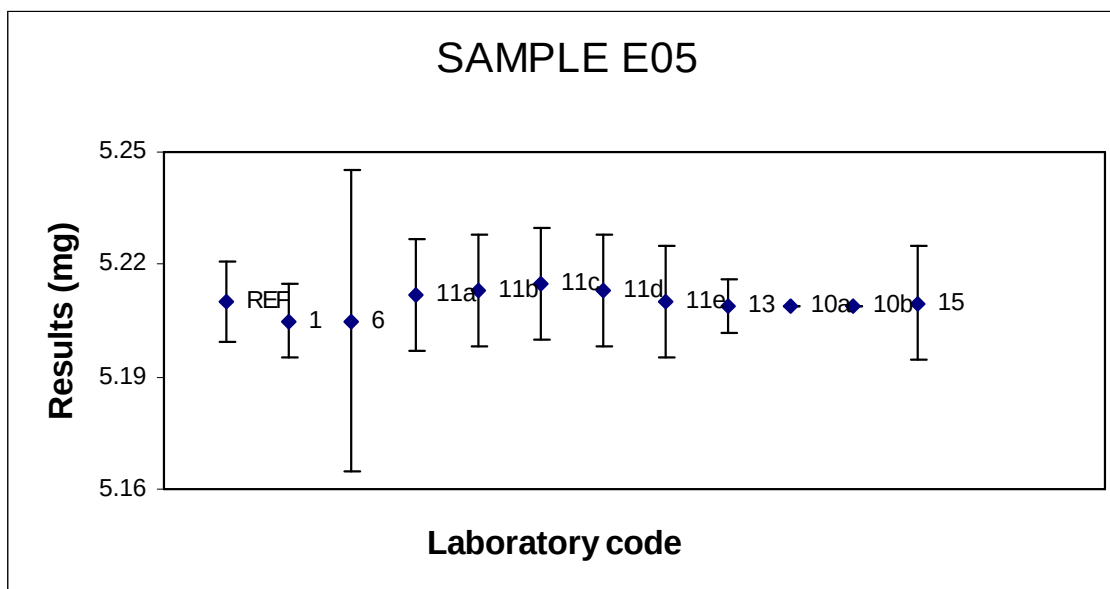
NOTES: 1. Results for laboratory code 9 extends well beyond chart limits.

2. Laboratory codes 4 and 9 nominated 5 place microbalances, and laboratory codes 14 and 19 nominated 6 place microbalances.

Graphical Presentation of Participant's Results

NOTE: Laboratory codes 5a, 5b, 5c and 10 nominated 5 place microbalances, and laboratory codes 15, 18 and 20 nominated 6 place microbalances.

Graphical Presentation of Participant's Results



- NOTES: 1. Uncertainty bars for laboratory codes 16a and 16b are not shown on above charts.
2. Laboratory codes 2 and 7 nominated 5 place microbalances, and laboratory codes 12a, 12b and 17 nominated 6 place microbalances.

Appendix A8

ADDITIONAL INFORMATION (As Reported by Participants)

Lab Code	Sample Set	Date Analysed	Nominated 5/6 points	Brand	Microbalance Model	Range
1	E	5/05/2009	5	AND	GH-252	wg 0.00
2	A	11/11/2008	5	Mettler Toledo	AB 265-S	Delta: 0.01/0.1mg
3a	B	5/11/2008	5	Mettler	ME30	0mg-30mg
3b	B	5/11/2008	5	Mettler Toledo	XS 105 Dual Range	0.01mg
3c	B	5/11/2008	5	Mettler Toledo	ME30	0.001mg
4	C	14/05/2009	5	Sartorius	BP211D	0-80grams
5a	D	19/11/2008	5	Mettler Toledo	AG 285	0-200g
5b	D	18/11/2008	5	Mettler Toledo	AG 285	0-200g
5c	D	3/11/2008	5	Mettler Toledo	AG 285	0-200g
6	E	6/11/2008	5	AND	GR-202	0.00mg
7	A	1/12/2008	5	AND	GR-202	5 decimal places
8	B	26/11/2008	5	Mettler Toledo	XS 105 DU	0.01-41g
9	C	23/02/2009	5	Mettler Toledo	XS 205 Dualrange	0.01mg/0.1 mg
10	D	15/12/2008	5	Ohaus	DV215CD	81/210g
11a	E	20/11/2008	6	Mettler	MT5	N/A
11b	E	25/11/2008	6	Mettler	MT5	N/A
11c	E	21/11/2008	6	Mettler	MT5	N/A
11d	E	26/11/2008	6	Mettler	MT5	N/A
11e	E	25/11/2008	6	Mettler	MT5	N/A
12a	A	27/01/2009	6	Ohaus	DV215CD	5 DECIMALS
12b	A	27/01/2009	6	Ohaus	DV215CD	0/01/1900
13	E	26/02/2009	6	Cahn 26	Cahn 26	50mg
14	C	28/04/2009	6	Mettler	MT5	1µg
15	D	22/01/2009	6	Cahn	C-31 Microbalance	25mg range
16a	E	12/12/2008	6	Mettler Toledo	MX5	0.000-5000.000mg
16b	E	10/12/2008	6	Mettler Toledo	MX5	0.000-5000.000mg
17	A	16/02/2009	6	Mettler	M3	0-100mg
18	D	18/03/2009	6	Mettler	M3	0-150
19	C	20/03/2009	6	Mettler	M3	0-150
20	D	23/02/2009	6	Mettler	M3	0/01/1900
21	E	10/02/2009	6	Mettler Toledo	MX5	6 decimal place

NOTE:

- Where a laboratory submitted results for more than one analyst, an 'a', 'b' etc has been added to the numerical code to denote the different analyst.

APPENDIX B

Instructions to Participants and Results Sheet



PROFICIENCY TESTING AUSTRALIA

Proficiency Testing Program

Gravimetric Round 1

INSTRUCTIONS TO PARTICIPANTS

Please read instructions carefully **BEFORE** commencing testing. To ensure that the results of this program can be analysed properly, participants are asked to carefully note the following:

- 1 Enclosed are two aluminium weights in separate labelled containers. They are nominally 5mg and 20 mg respectively. Plastic tweezers are also enclosed.
- 2 Each participant is requested to do the following:
 - (i) Determine the weight of each specimen using your laboratory's routine procedures (omit those parts of the procedures that relate to the use of blank filters).
 - (ii) Record your observations into your own laboratory system, and also on the attached Result Sheet.
 - (iii) The Uncertainty of Measurement (at 95% confidence level) *must* also be reported for each specimen.
 - (iv) If there is more than one analyst in your laboratory, each analyst is encouraged to participate (without comparison with others), and to submit results on separate copies of the attached Result Sheet.

3 IMPORTANT NOTES

- (i) Please use **only** the supplied plastic tweezers to handle the weights.
- (ii) Under **no** circumstances handle the weights with metal tweezers, or by hand.
- (iii) Do **not** use the weights in a 'dirty' environment.
- (iv) Only have one weight out of its container at any one time (to avoid mix-up).
- (v) Return result sheets, the weights and tweezers to PTA promptly (**please no later than Friday 10 October 2008**), properly packaged to:

Dr Michael Li, Proficiency Testing Australia.
PO Box 7507, Silverwater NSW 2128. Australia.
Phone: +61 2 9736 8397 Fax: +61 2 9743 6664
Email: michael.li@pta.asn.au

PROFICIENCY TESTING AUSTRALIA

Gravimetric Round 1

Results Sheet

Lab Code:

Sample Identification	Weight (mg)	Uncertainty (U_{95})* (mg)
<i>Nominal 5 mg</i>		
<i>Nominal 20 mg</i>		

*Note: U_{95} refers to your laboratory's uncertainty of measurement at the 95% confidence level.

Analyst's Name: _____

Date analysed: _____

Microbalance Brand: _____

Microbalance Model: _____

Microbalance Range used: _____

Signed: _____

Date: _____

Return no later than **Friday 10 October 2008** to:

Dr Michael Li, Proficiency Testing Australia.

P O Box 7507, Silverwater NSW 2128. Australia.

Phone: +61 2 9736 8397 Fax: +61 2 9743 6664Email: michael.li@pta.asn.au

---- End of Report ----